Ubuntu AD Integration

Text written in blue is optional - write it when you get error

1 sudo apt update
2 sudo apt upgrade
3 sudo apt install vim
4 sudo vim /etc/hosts

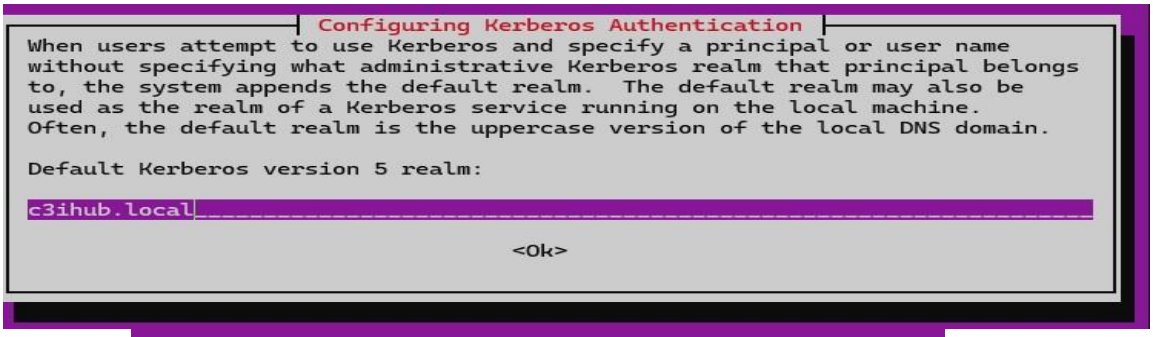
```
127.0.0.1      localhost
127.0.1.1      joe-machine
172.26.234.127  adserver.c3ihub.local c3ihub  or  172.26.234.127
adserver.c3ihub.local ad
```

5 sudo cat /etc/hosts
6 sudo vim /etc/systemd/resolved.conf

```
[Resolve]

DNS=172.26.234.127
Domains=c3ihub.local
```

7 sudo cat /etc/systemd/resolved.conf
8 sudo systemctl status systemd-resolved.service
9 ping 172.26.234.127
10 ping c3ihub.local
11 sudo apt update
12 sudo apt install -y realmd sssd sssd-tools adcli samba-common-bin packagekit krb5-user chrony



13 sudo realm discover c3ihub.local

14 sudo vim /etc/krb5.conf

```
[libdefaults]
    udp_preference_limit = 0
default_realm = C3IHUB.LOCAL
dns_lookup_realm = true
dns_lookup_kdc = true
[realms]
    C3IHUB.LOCAL = {
        #kdc = c3ihub.local
        #admin_server = c3ihub.local(it will auto show some time don't remove it or add it.)
    }
[domain_realm]
    .c3ihub.local = C3IHUB.LOCAL      c3ihub.local
= C3IHUB.LOCAL
```

15 sudo cat /etc/krb5.conf

16 sudo systemctl start sssd realmd chrony

17 date

18 sudo apt update

19 chronyc tracking

```
joe@joe-machine:~$ chronyc tracking
Reference ID    : 6752D0A6 (103.82.208.166)
Stratum         : 2
Ref time (UTC)  : Tue Oct 14 06:30:37 2025
System time     : 0.000051688 seconds fast of NTP time
Last offset     : -0.000148756 seconds
RMS offset      : 0.002260586 seconds
Frequency       : 16.193 ppm slow
Residual freq   : -0.021 ppm
Skew            : 3.481 ppm
Root delay      : 0.044852562 seconds
Root dispersion : 0.002057237 seconds
Update interval : 64.4 seconds
Leap status     : Normal
joe@joe-machine:~$
```

20 sudo realm -v join --user=resolve_ad01 c3ihub.local (also provide administrator password)

21 sudo systemctl restart sssd.service

22 sudo systemctl status sssd

```
joe@joe-machine:~$ sudo systemctl restart sssd
joe@joe-machine:~$ sudo systemctl status sssd
● sssd.service - System Security Services Daemon
     Loaded: loaded (/lib/systemd/system/sssd.service; enabled; vendor preset: enabled)
     Active: active (running) since Tue 2025-10-14 12:02:46 IST; 9s ago
   Main PID: 5939 (sssd)
      Tasks: 4 (limit: 4542)
     Memory: 46.1M
        CPU: 169ms
     CGroup: /system.slice/sssd.service
             ├─5939 /usr/sbin/sssd -i --logger=files
             ├─5941 /usr/libexec/sssd/sssd_be --domain c3ihub.local --uid 0 --gid 0 --logger=files
             ├─5942 /usr/libexec/sssd/sssd_nss --uid 0 --gid 0 --logger=files
             └─5943 /usr/libexec/sssd/sssd_pam --uid 0 --gid 0 --logger=files

Oct 14 12:02:46 joe-machine systemd[1]: Starting System Security Services Daemon...
Oct 14 12:02:46 joe-machine sssd[5939]: Starting up
Oct 14 12:02:46 joe-machine sssd_be[5941]: Starting up
Oct 14 12:02:46 joe-machine sssd_nss[5942]: Starting up
Oct 14 12:02:46 joe-machine sssd_pam[5943]: Starting up
Oct 14 12:02:46 joe-machine systemd[1]: Started System Security Services Daemon.
joe@joe-machine:~$
```

23 getent passwd

24 id asthak@c3ihub.local

```
joe@joe-machine:~$ id asthak@c3ihub.local
```

uid=351601541(asthak@c3ihub.local) gid=351600513(domain users@c3ihub.local) groups=351600513(domain users@c3ihub.local),351601389(infra_sec_grp@c3ihub.local)

25 sudo su asthak@c3ihub.local

26 Sudo visio

```
joe@joe-machine:~$ sudo su asthak@c3ihub.local
asthak@c3ihub.local@joe-machine:/home/joe$
```

enter this= joe ALL=(ALL:ALL) NOPASSWD: /usr/bin/apt update, /usr/bin/apt upgrade

27 su – joe
28 sudo apt update
29 sudo apt upgrade
30 sudo -i

END

Remove Ubuntu from AD Domain

1. sudo realm list
2. sudo realm leave c3ihub.local -v
3. sudo realm list
4. sudo systemctl stop sssd
5. sudo rm -rf /var/lib/sss/db/*
6. sudo rm -rf /etc/sssd/sssd.conf
7. sudo kdestroy
8. sudo rm -f /etc/krb5.keytab
9. sudo apt remove --purge realmd sssd adcli oddjob oddjob-mkhomedir sambacommon-bin -y
10. sudo apt autoremove -y
11. sudo reboot
12. getent passwd | grep c3ihub
13. Remove entry

Sudo vim /etc/hosts

Sudo vim /etc/systemd/resolved.conf

Sudo vim /etc/krb5.conf

```
# User privilege specification
root    ALL=(ALL:ALL) ALL
joe     ALL=(ALL:ALL) NOPASSWD: /usr/bin/apt update, /usr/bin/apt upgrade
# Members of the admin group may gain root privileges
%admin ALL=(ALL) ALL

^G Help        ^O Write Out   ^W Where Is    ^K Cut         ^T Execute
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify
```

END

Important Checks

Validate Kerberos=

#kinit asthak@C3IHUB.LOCAL

Then enter the AD user password. Expect no output (success).

Check give PAM Permission
sudo nano /etc/sssd/sssd.conf

simple_allow_users = administrator@c3ihub.local, asthak@c3ihub.local, add more adminusers if want

Check user is in sudoers file
sudo visudo

"asthak@c3ihub.local" ALL=(ALL:ALL) ALL

Automated Script

A script is given below for fast installation

Save this script as script.sh

Give full permission for execute

Change details as needed necessary are username, password and DNS IP

HOSTNAME="DHEA01229.c3ihub.local"
AD_DOMAIN="c3ihub.local"
AD_USER="quantamed"
DNS_IP="172.26.234.127"
ALLOWED_USERS="quantamed@c3ihub.local, administrator@c3ihub.local, asthak@c3ihub.local"
SUDO_USERS=quantamed@c3ihub.local

Also provide administrator password to authenticate.

After Complete reboot the system and login again. Also check all configuration manually for reconfirm that.

```
#!/bin/bash
set -e

##############################################
# Configuration Section (Edit as needed)
##############################################

HOSTNAME="DHEA01229.c3ihub.local"
AD_DOMAIN="c3ihub.local"
AD_USER="quantamed"
DNS_IP="172.26.234.127"
ALLOWED_USERS="quantamed@c3ihub.local,
administrator@c3ihub.local, asthak@c3ihub.local"
SUDO_USERS= quantamed@c3ihub.local

##############################################
# Start Setup
##############################################

echo "[*] Updating system packages..."
sudo apt update -y

echo "[*] Installing basic packages..."
sudo apt install -y openssh-server net-tools

echo "[*] Starting SSH service..."
sudo systemctl start ssh

echo "[*] Showing IP information:"
ifconfig || true

echo "[*] Installing AD and realm join packages..."
sudo apt install -y realmd libnss-sss libpam-sss sssd sssd-tools adcli
samba-common-bin oddjob oddjob-mkhomedir packagekit

echo "[*] Pinging AD DNS server ($DNS_IP)..."
ping -c 3 "$DNS_IP" || echo "Warning: DNS server not reachable!"

echo "[*] Updating again (post network config)..."
sudo apt update -y

echo "[*] Setting hostname to $HOSTNAME"
sudo hostnamectl set-hostname "$HOSTNAME"

echo "[*] Disabling systemd-resolved (if interfering)..."
sudo systemctl disable systemd-resolved.service
sudo systemctl stop systemd-resolved.service

echo "[*] Updating /etc/hosts with hostname..."
sudo sed -i "/127.0.1.1/d" /etc/hosts
echo "127.0.1.1 $HOSTNAME" | sudo tee -a /etc/hosts

echo "[*] Configuring DNS in /etc/systemd/resolved.conf..."
sudo bash -c "cat >/etc/systemd/resolved.conf" <<EOF
[Resolve]
DNS=$DNS_IP
FallbackDNS=8.8.8.8
Domains=$AD_DOMAIN
EOF

echo "[*] Restarting systemd-resolved and fixing resolv.conf..."
sudo systemctl restart systemd-resolved
sudo ln -sf /run/systemd/resolve/resolv.conf /etc/resolv.conf

echo "[*] Discovering realm ($AD_DOMAIN)..."
sudo realm discover "$AD_DOMAIN"

echo "[*] Joining AD domain..."
sudo realm join --user="$AD_USER" "$AD_DOMAIN"

echo "[*] Verifying realm join..."
realm list

echo "[*] Permitting AD user: $AD_USER@$AD_DOMAIN"
sudo realm permit "$AD_USER@$AD_DOMAIN"

echo "[*] Checking sssd status..."
systemctl status sssd || true

echo "[*] Ensuring home dir creation is enabled..."
if ! grep -q pam_mkhomedir.so /etc/pam.d/common-session; then
   echo 'session required pam_mkhomedir.so skel=/etc/skel
umask=0022' | sudo tee -a /etc/pam.d/common-session
fi

echo "[*] Restarting sssd..."
sudo systemctl restart sssd

echo "[*] Testing AD user resolution (optional)..."
getent passwd 'asthak@c3ihub.local' || echo "User
'asthak@c3ihub.local' not resolved — may not exist or replicated yet."

echo "[*] Updating /etc/sssd/sssd.conf with allowed users..."
sudo bash -c "echo 'simple_allow_users = $ALLOWED_USERS' >>
/etc/sssd/sssd.conf"
sudo systemctl restart sssd

echo "[*] Adding sudoers entry for users..."
for user in $SUDO_USERS; do
   echo "$user ALL=(ALL) NOPASSWD: /usr/bin/apt update,
/usr/bin/apt upgrade" | sudo tee -a /etc/sudoers.d/ad_users
done

echo "[✓] AD Integration complete."
```